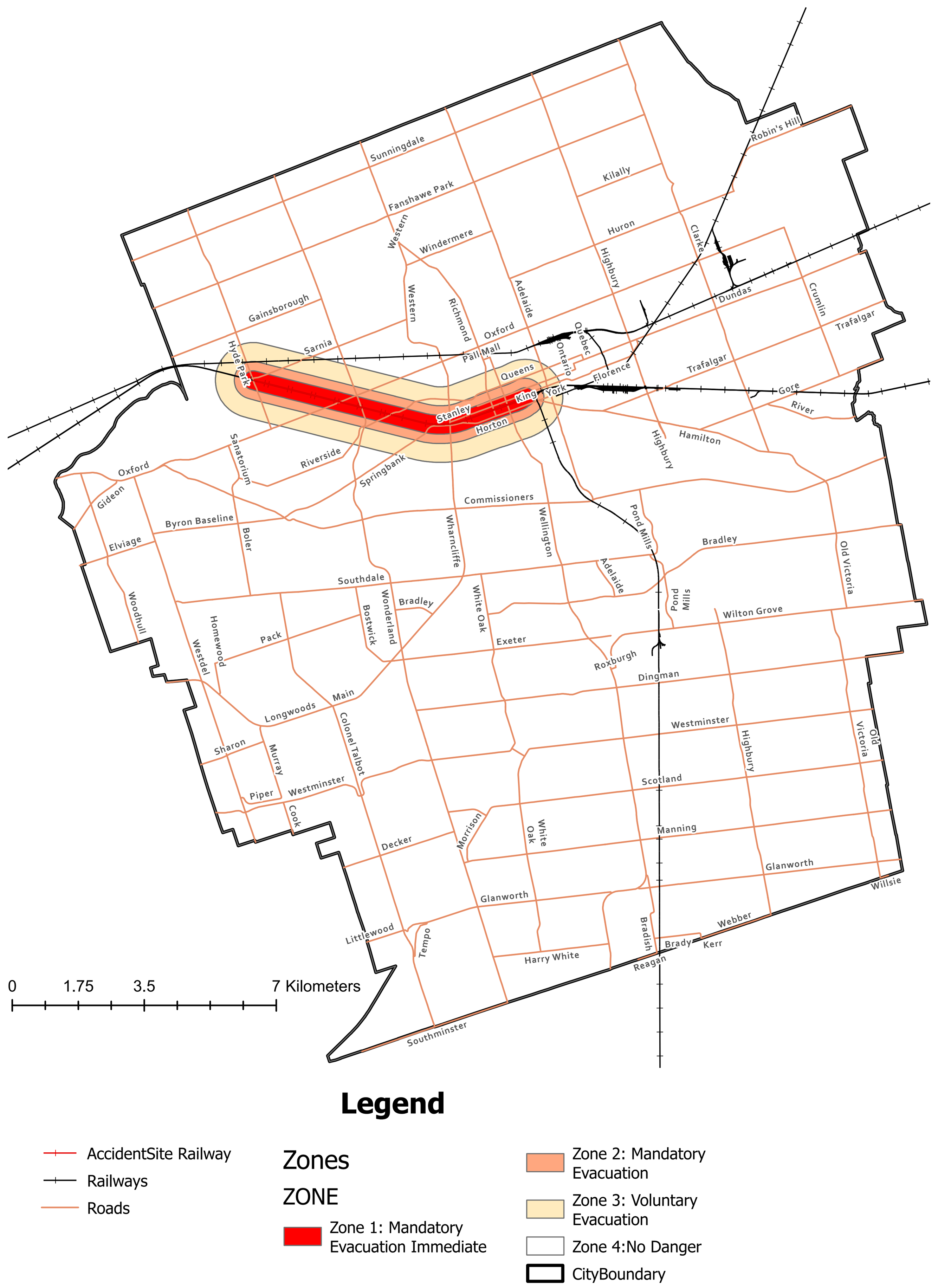
Evacuation Planning for Rail Derailment in London City

This project focuses on evacuation planning in response to a train derailment scenario using ArcGIS Pro and Model Builder. The goal is to allocate evacuees from the immediate danger zone to nearby shelters while modeling transportation disruptions caused by the incident. Two key scenarios were developed: one for allocating the population to safe shelters and another for assessing travel route disruptions. The project utilizes GIS tools and spatial analysis to create maps that support public safety and emergency management efforts.

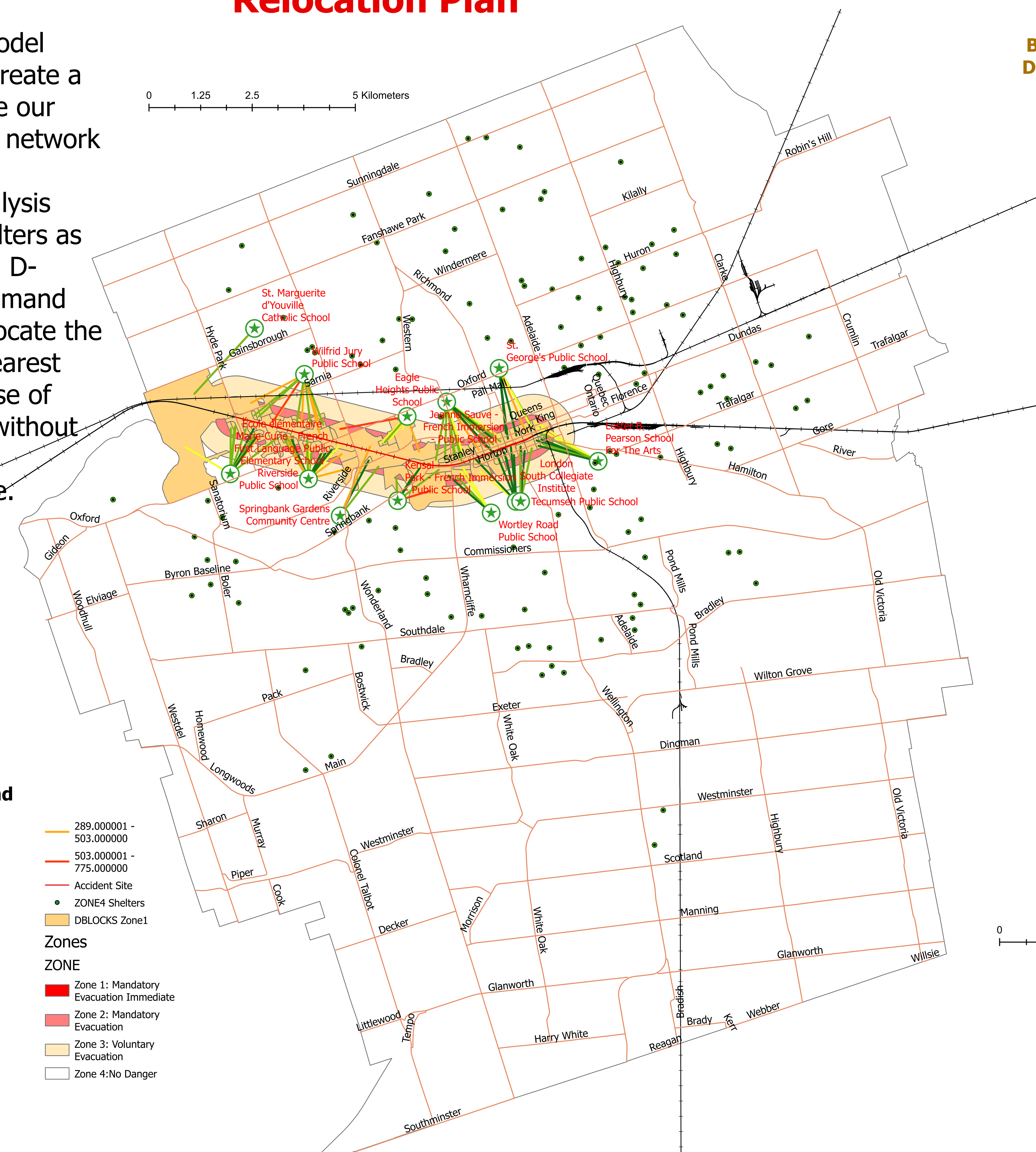
Railway Derailment Area by Zones

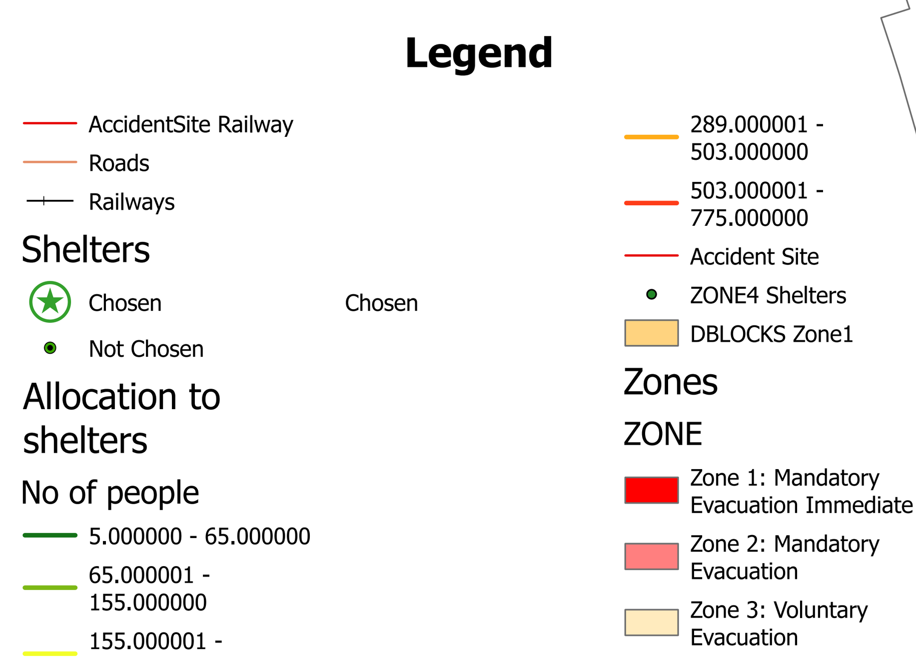
Using the model builder we build a model to buffer the zones around the accident site which helps to categorize the evacuation, below is a map showing the accident site and zones of evacuation around it.



Relocation Plan

Using the Model Builder we create a model to use our roads at the network and perform network analysis and use shelters as facilities and D-Blocks as demand points to allocate the people to nearest facility in case of evacuation without crossing the Accident Site.

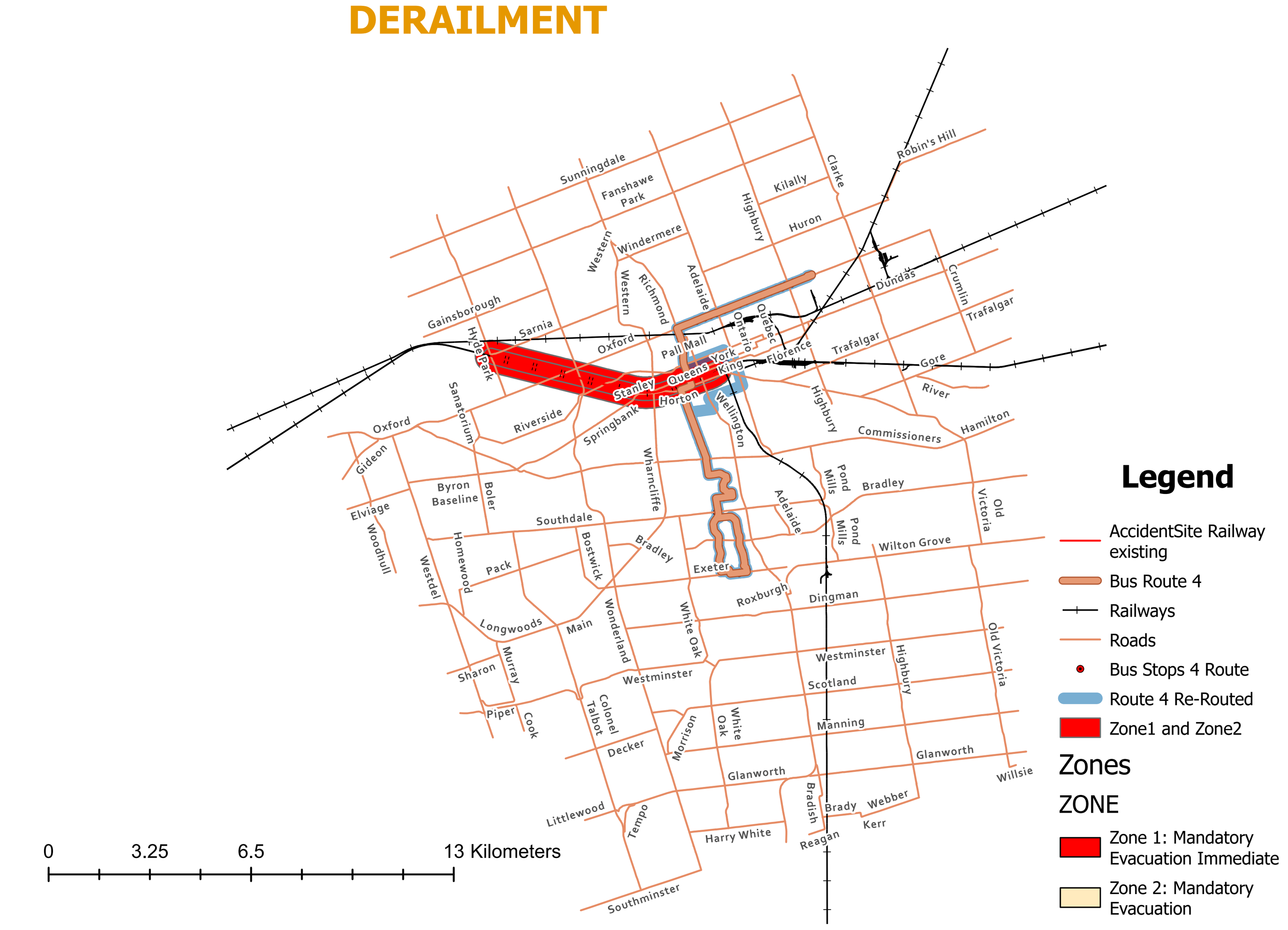




RE- ROUTING THE BUS ROUTE 4

BEFORE THE DERAILMENT

AFTER THE DERAILMENT



Using network analysis and our roads as a network, we build a network and, through route analysis, identify the optimized re-routing of the bus route.

The analysis successfully identified optimal shelter locations for evacuees in Zone 1 without crossing restricted areas. Routes for public transit were effectively reconfigured to account for barriers, ensuring continuity of service. The maps provide clear visual insights into population allocation and travel route adjustments. This project demonstrates the utility of GIS in emergency planning and offers a scalable approach for future public safety scenarios.

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